Keyword Ensemble Metadata for Reducing Retrieval Noise in RAG Pipelines

An Empirical Study on BEIR Datasets

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Abstract

Retrieval-Augmented Generation (RAG) systems rely heavily on the quality of retrieved documents. However, retrieval models often return both relevant and irrelevant documents within the top-k results, introducing retrieval noise that can degrade downstream generation quality.

This paper investigates whether keyword metadata extracted from documents can serve as an auxiliary signal for reducing retrieval noise in RAG pipelines. We construct keyword metadata using unsupervised extraction algorithms such as RAKE and YAKE and integrate these signals into a BM25 retrieval framework through score fusion and candidate re-ranking strategies.

Experiments on BEIR benchmark datasets (SciFact, NFCorpus, and Arguana) show that small keyword weights ($\alpha \approx 0.05$–0.1) provide the most stable improvements while preserving the robustness of lexical retrieval.

Our findings suggest that keyword metadata functions best as a lightweight auxiliary retrieval signal rather than a primary ranking mechanism.

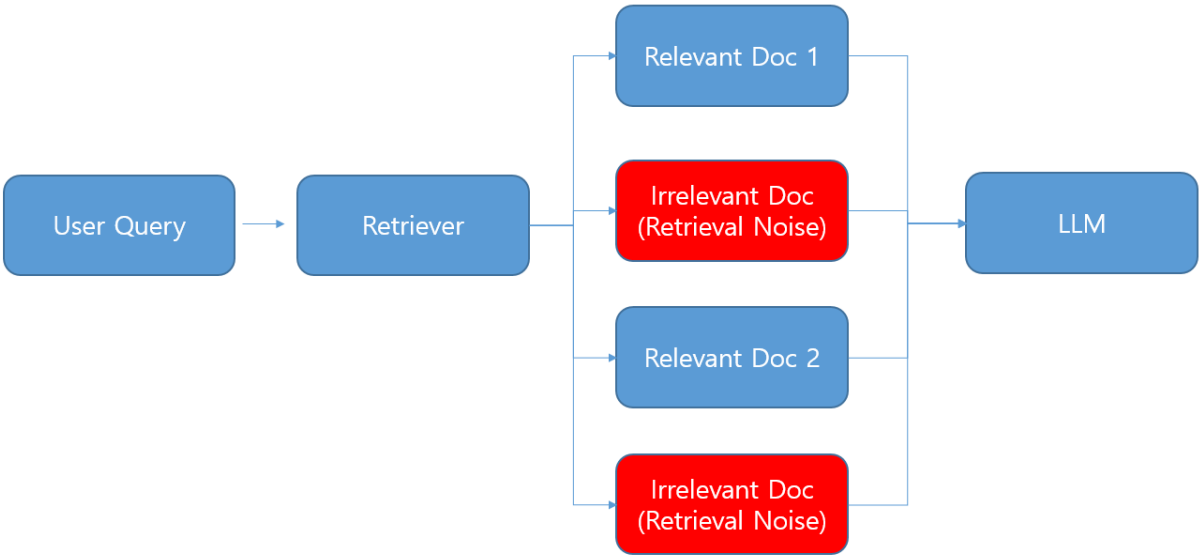

Figure 1: Retrieval noise in a typical RAG pipeline.

A retriever may return both relevant and irrelevant documents within the top-k results,

introducing noise into the context used by the language model.

1 Introduction

Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. (2020) as a framework that combines parametric language models with non-parametric retrieval systems. Since its introduction, RAG has become a widely adopted architecture for integrating external knowledge into large language models.

By retrieving relevant documents from a knowledge corpus and providing them as context to the model, RAG systems can improve factual accuracy

and reduce hallucination compared to standalone language models.

Despite these advantages, the effectiveness of RAG pipelines heavily depends on the quality of retrieved documents.

Traditional retrieval methods such as BM25 or dense retrieval models often return a mixture of relevant and irrelevant documents within the top-k results.

These irrelevant documents introduce retrieval noise, which can degrade the quality of downstream generation by polluting the context provided to the language model.

Figure 1 illustrates a typical scenario in which a retriever returns both relevant and irrelevant documents for a user query.

Even when relevant documents are included, the presence of irrelevant documents can reduce the signal-to-noise ratio in the context window,

potentially leading the language model to produce less accurate responses.

To mitigate this issue, recent work has explored hybrid retrieval strategies that incorporate additional signals beyond lexical similarity.

One lightweight approach is the use of keyword-based metadata extracted from documents.

Keyword extraction algorithms such as RAKE and YAKE can capture salient terms that represent document semantics in a compact form.

In this work, we investigate whether keyword ensemble metadata can serve as an auxiliary retrieval signal in RAG pipelines.

Rather than replacing traditional lexical retrieval methods, keyword metadata is used as a complementary signal to guide ranking decisions

and potentially reduce retrieval noise.

Specifically, we study the following research question:

Can keyword ensemble metadata reduce retrieval noise in RAG pipelines?

To answer this question, we evaluate a keyword metadata retrieval framework using BEIR

benchmark datasets, including SciFact, NFCorpus, and Arguana.

We analyze keyword representation quality, retrieval performance, reranking behavior, and parameter sensitivity through an alpha-weighted fusion

between BM25 scores and keyword overlap signals.

Our experimental results show that keyword metadata functions as a weak but useful auxiliary signal.

While it does not replace lexical retrieval signals, it can provide modest improvements in certain datasets and retrieval scenarios,

particularly in argument retrieval tasks.

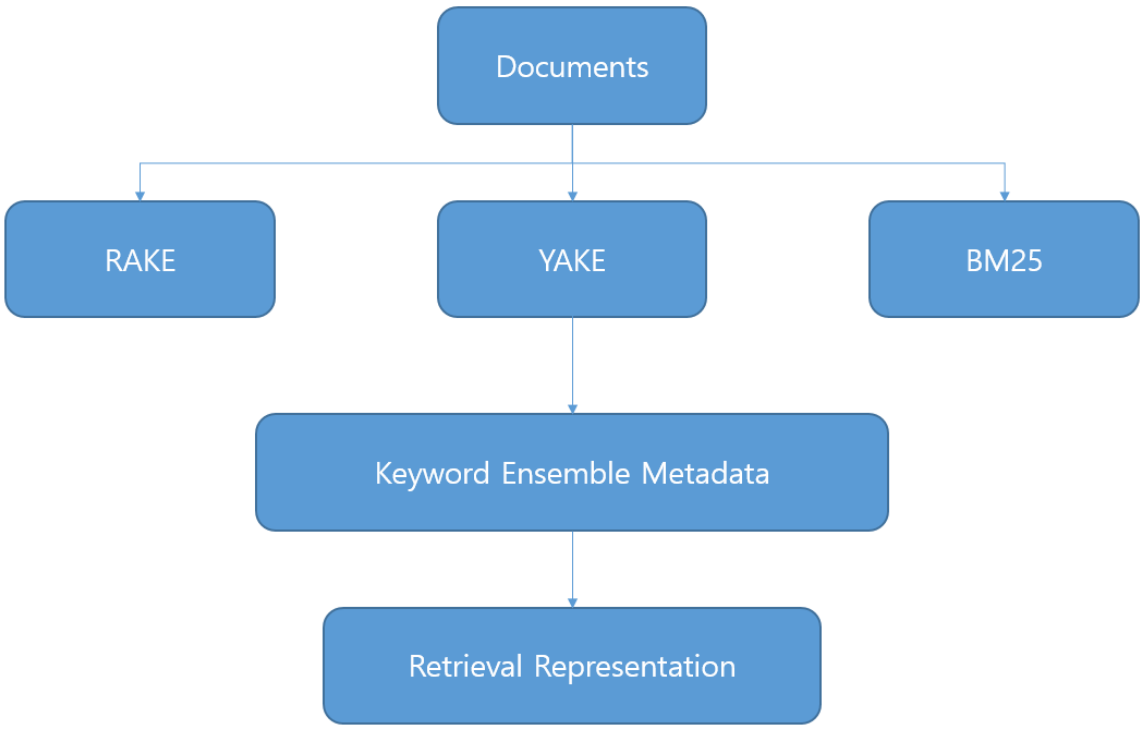

Figure 2: Keyword metadata retrieval pipeline.

Documents are processed using keyword extraction algorithms such as RAKE or YAKE to generate metadata representations.

These keyword signals are then combined with BM25 retrieval and used for candidate re-ranking.

2 Related Work

Recent surveys (Gao et al., 2023) categorize RAG architectures into several design patterns and highlight the importance of retrieval quality for downstream generation.

BM25 remains one of the strongest lexical retrieval baselines in information retrieval systems (Robertson and Zaragoza, 2009).

The BEIR benchmark introduced by Thakur et al. (2021) provides a standardized evaluation framework across diverse retrieval tasks.

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for integrating external knowledge into large language models.

By retrieving relevant documents and providing them as context to the language model, RAG systems can improve factual grounding

and reduce hallucination. Recent research has focused on improving the retrieval stage of RAG pipelines, including hybrid retrieval,

dense retrieval, and reranking strategies.

Traditional lexical retrieval methods such as BM25 remain strong baselines for information retrieval tasks.

BM25 relies on term frequency and inverse document frequency statistics to rank documents based on lexical similarity to the query.

Despite the rise of neural retrieval methods, BM25 continues to demonstrate competitive performance across many benchmark datasets.

Another line of research focuses on enhancing retrieval quality through hybrid retrieval approaches.

These methods combine multiple signals, such as lexical similarity, dense embedding similarity, or structural metadata,

to improve ranking performance. Hybrid retrieval methods aim to leverage complementary signals that capture different aspects

of document relevance.

Keyword extraction methods provide an alternative lightweight representation of document semantics.

Algorithms such as RAKE (Rapid Automatic Keyword Extraction) and YAKE (Yet Another Keyword Extractor) identify salient terms

or phrases that summarize the core topics of a document. These keywords can serve as compact metadata representations that

capture important semantic signals.

While keyword extraction has been widely studied for document summarization and indexing, its role in retrieval pipelines

has received relatively limited attention. In particular, it remains unclear whether keyword metadata can improve retrieval

quality when used as an auxiliary signal alongside lexical retrieval methods.

This work investigates the integration of keyword ensemble metadata into retrieval pipelines and evaluates its impact on

retrieval performance and noise reduction in RAG systems.

3 Method

This section describes the keyword metadata retrieval framework used in our experiments.

The core idea is to augment traditional lexical retrieval with keyword-based metadata signals

extracted from documents.

3.1 Keyword Metadata Extraction

RAKE (Rapid Automatic Keyword Extraction) (Rose et al., 2010) identifies candidate phrases by removing stopwords and scoring them based on word co-occurrence statistics.

YAKE (Campos et al., 2020) extracts keywords using local statistical features such as term frequency, position, and context dispersion.

For each document in the corpus, we extract keywords using unsupervised keyword extraction algorithms.

Specifically, we use two widely adopted methods:

- RAKE (Rapid Automatic Keyword Extraction)
- YAKE (Yet Another Keyword Extractor)

These algorithms identify salient words or phrases that represent the main topics of a document.

The extracted keywords are stored as metadata associated with each document.

This keyword metadata provides a compact semantic representation that can be used as an auxiliary

signal during retrieval.

3.2 Retrieval Framework

Figure 2 illustrates the overall retrieval pipeline. The process consists of three main stages.

First, keyword extraction is performed offline on the document corpus to generate keyword metadata.

This metadata is stored alongside the original document index.

Second, document retrieval is performed using a BM25 lexical retriever.

BM25 generates an initial ranking of candidate documents based on query–document term matching.

Third, the retrieved candidate documents are re-ranked using a keyword overlap signal.

This signal measures the degree of overlap between query terms and keyword metadata associated

with each document.

3.3 Score Fusion

To combine lexical and keyword signals, we use a weighted score fusion strategy:

$$\text{score}(d, q) = (1 - \alpha) \times \text{BM25}(d, q) + \alpha \times \text{KW}(d, q)$$

$$\text{KW}(d, q) = |\text{terms}(q) \cap \text{keywords}(d)| \,/\, |\text{keywords}(d)|$$

where: BM25(d, q) represents the lexical relevance score between query q and document d,

KW(d, q) represents the keyword overlap score between the query and document metadata,

α controls the relative weight of the keyword signal.

When α = 0, retrieval relies purely on BM25.

As α increases, keyword metadata contributes more strongly to the ranking.

3.4 Candidate Re-ranking

In addition to direct score fusion, we evaluate a candidate re-ranking strategy.

In this approach, BM25 first retrieves the top-k candidate documents.

Keyword overlap scores are then computed within this candidate set and used to re-rank the documents.

This strategy allows keyword metadata to influence ranking decisions while preserving the strong lexical matching properties of BM25.

4 Experimental Setup

We evaluate our approach using datasets from the BEIR benchmark (Thakur et al., 2021), which provides a standardized framework for evaluating retrieval models across diverse domains.

4.1 Datasets

We evaluate our approach using datasets from the BEIR benchmark (Thakur et al., 2021), which provides a standardized framework for evaluating information retrieval systems across diverse domains.

We conduct experiments on three datasets representing different retrieval scenarios:

- SciFact
- NFCorpus
- Arguana

SciFact is a scientific fact verification dataset consisting of scientific claims paired with

supporting or refuting evidence documents. The dataset evaluates whether retrieval systems can identify relevant scientific papers that support or contradict a given claim.

NFCorpus is a medical information retrieval dataset containing real-world user queries and expert-written medical documents. It represents a question–answer retrieval scenario where the goal is to retrieve relevant medical explanations for user questions.

Arguana is an argument retrieval dataset where documents represent argumentative statements. Queries correspond to arguments that should retrieve counterarguments or related argumentative texts. This dataset emphasizes semantic relevance and discourse-level matching.

These datasets provide diverse retrieval environments that allow us to evaluate the robustness of keyword metadata signals across different domains and retrieval tasks.

4.2 Retrieval Configurations

We evaluate several retrieval configurations to analyze the effect of keyword metadata on retrieval performance.

- BM25 (baseline lexical retriever)

- RAKE + BM25

BM25 retrieval combined with RAKE-based keyword metadata signals.

- YAKE + BM25

BM25 retrieval combined with YAKE-based keyword metadata signals.

- Keyword Ensemble

A combined keyword metadata representation constructed from RAKE and YAKE outputs.

4.3 Baseline Retriever

Our baseline retriever is BM25, a widely used lexical retrieval algorithm that ranks documents based on term frequency and inverse document frequency statistics.

BM25 is used to generate the initial ranking of candidate documents for each query.

4.4 Keyword Metadata

Keyword metadata is generated using two unsupervised keyword extraction algorithms:

- RAKE (Rapid Automatic Keyword Extraction)
- YAKE (Yet Another Keyword Extractor)

For each document, keywords extracted by these algorithms are stored as metadata representations.

4.5 Evaluation Metrics

Retrieval performance is evaluated using standard information retrieval metrics:

- NDCG@10
- MRR (Mean Reciprocal Rank)
- Recall@10

In addition, we analyze the sensitivity of the retrieval system to the keyword weight parameter

α.

The combined retrieval score is defined as:

$$score(d,q) = (1 - \alpha) \times BM25(d,q) + \alpha \times KW(d,q)$$

where KW(d,q) represents the keyword overlap score between query terms and document keyword metadata.

5 Results

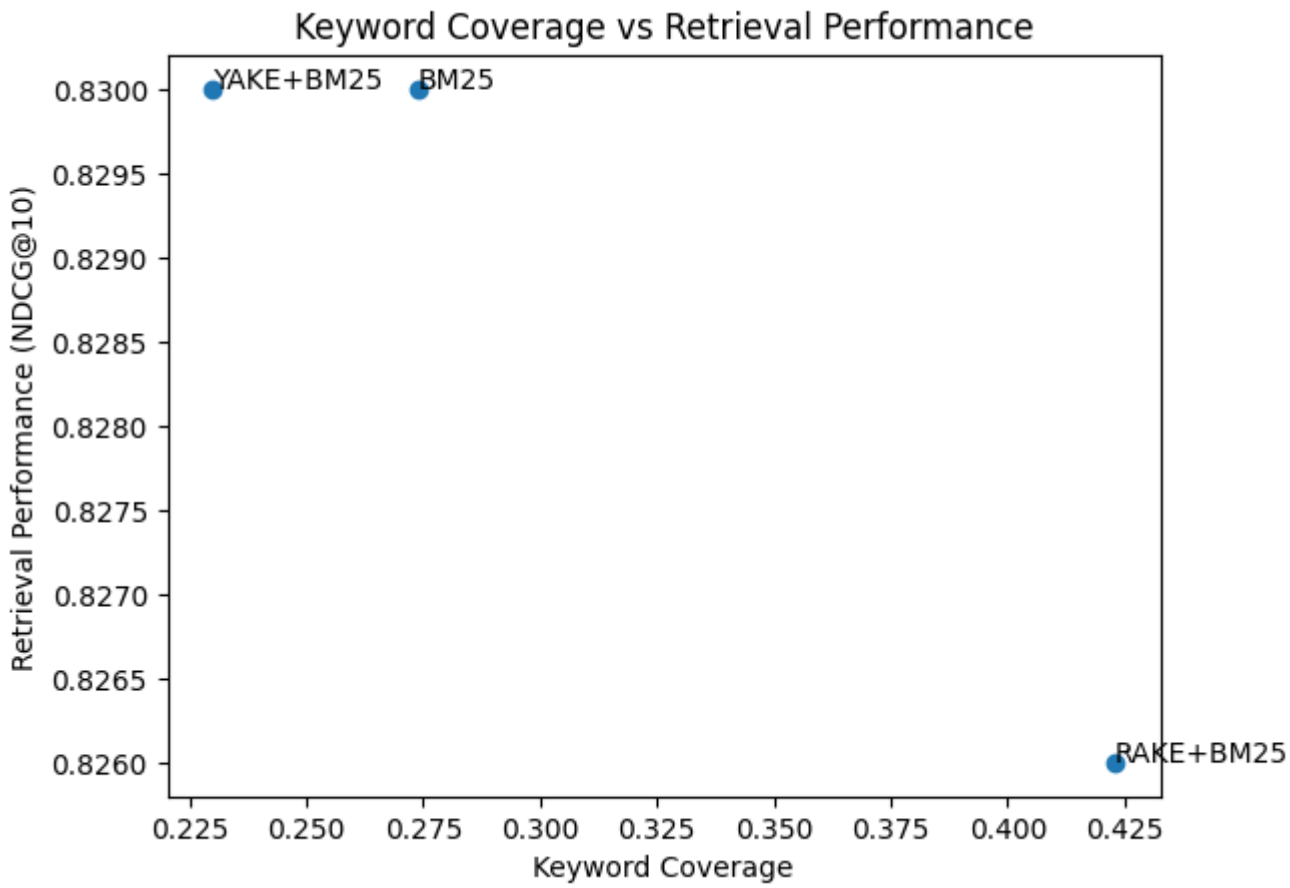

Figure 3: Relationship between keyword coverage and retrieval performance.

Although keyword extraction methods such as RAKE increase coverage of document terms,

this does not necessarily translate into higher retrieval performance.

The results suggest that keyword metadata provides complementary signals rather than replacing lexical retrieval methods.

Figure 3 illustrates the relationship between keyword coverage and retrieval performance.

RAKE-based keyword extraction produces higher keyword coverage compared to other methods,

indicating that more document terms are captured as metadata.

However, higher keyword coverage does not necessarily lead to better retrieval performance.

Despite capturing more keywords, RAKE-based methods achieve slightly lower NDCG@10 compared

to the BM25 baseline in the SciFact dataset.

This observation suggests that keyword coverage alone is not a sufficient indicator of

retrieval quality. Instead, keyword metadata should be interpreted as an auxiliary signal

that complements lexical retrieval methods rather than replacing them.

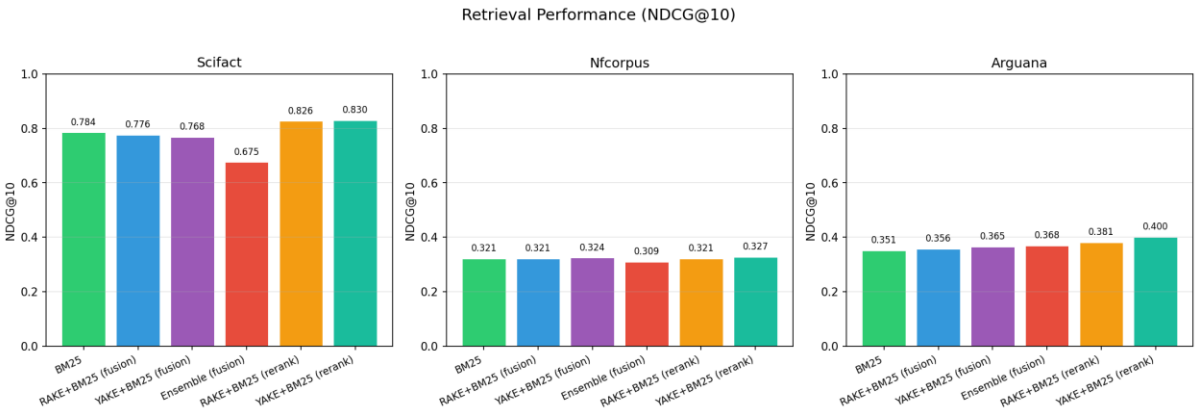

Figure 4: Retrieval performance (NDCG@10) across BEIR datasets using different keyword-enhanced retrieval strategies.

The results compare baseline BM25 with keyword fusion and keyword-based reranking methods.

Keyword reranking shows consistent improvements in some datasets, particularly in the Arguana argument retrieval task.

Figure 4 presents the retrieval performance (NDCG@10) across three BEIR datasets: SciFact, NFCorpus, and Arguana.

We compare the baseline BM25 retriever with several keyword-enhanced retrieval configurations, including score fusion

and candidate reranking strategies using RAKE and YAKE keyword metadata.

Overall, the results show that keyword-based score fusion does not consistently outperform the BM25 baseline.

In the SciFact dataset, the fusion approaches slightly reduce retrieval performance compared to BM25.

This suggests that naive integration of keyword signals may introduce additional ranking noise when combined directly

with lexical similarity scores.

In contrast, the candidate reranking strategy demonstrates more stable behavior.

By restricting keyword influence to the BM25 candidate set, reranking preserves the strong lexical matching properties

of BM25 while allowing keyword metadata to provide auxiliary ranking signals.

The Arguana dataset shows the most noticeable improvement from keyword-based reranking,

where the YAKE+BM25 rerank configuration achieves the highest NDCG@10 score.

This suggests that keyword metadata may be particularly beneficial in argument retrieval scenarios,

where salient terms and key phrases play a stronger role in representing document relevance.

Across datasets, these results indicate that keyword metadata functions best as a lightweight

auxiliary signal

rather than a primary ranking mechanism.

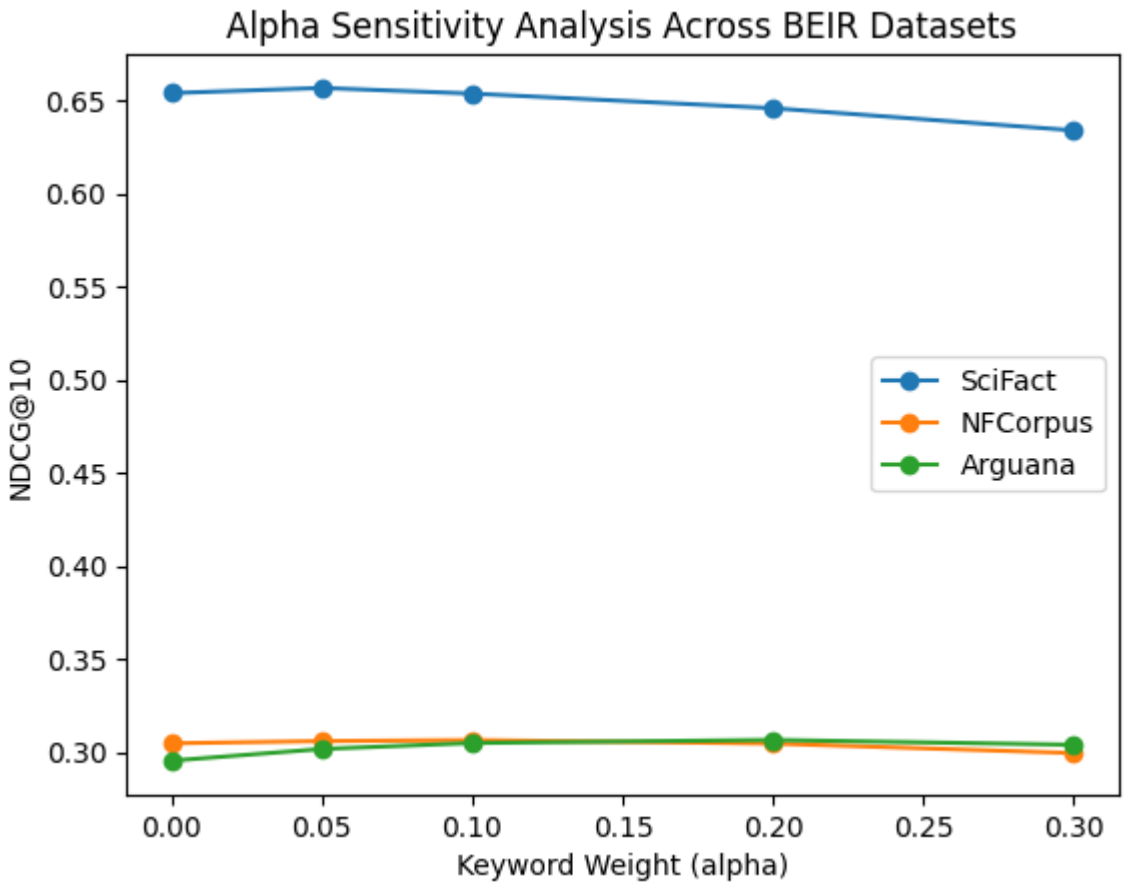

Figure 5: Alpha sensitivity analysis across BEIR datasets.

The parameter α controls the contribution of keyword metadata in the retrieval score fusion.

Figure 5 shows the alpha sensitivity analysis across three BEIR datasets.

Across all datasets, small keyword weights provide the most stable performance improvements.

When α becomes too large, retrieval performance tends to degrade because keyword overlap alone cannot fully capture document relevance.

For the SciFact dataset, the best performance is achieved around $\alpha \approx 0.05$, while NFCorpus shows slightly improved performance near $\alpha \approx 0.1$.

The Arguana dataset demonstrates stronger improvements as α increases, with optimal performance

observed near $\alpha \approx 0.2$.

These results indicate that keyword metadata acts as a weak auxiliary signal.

Rather than replacing lexical retrieval methods, keyword signals are most effective when used as a complementary feature with small weights.

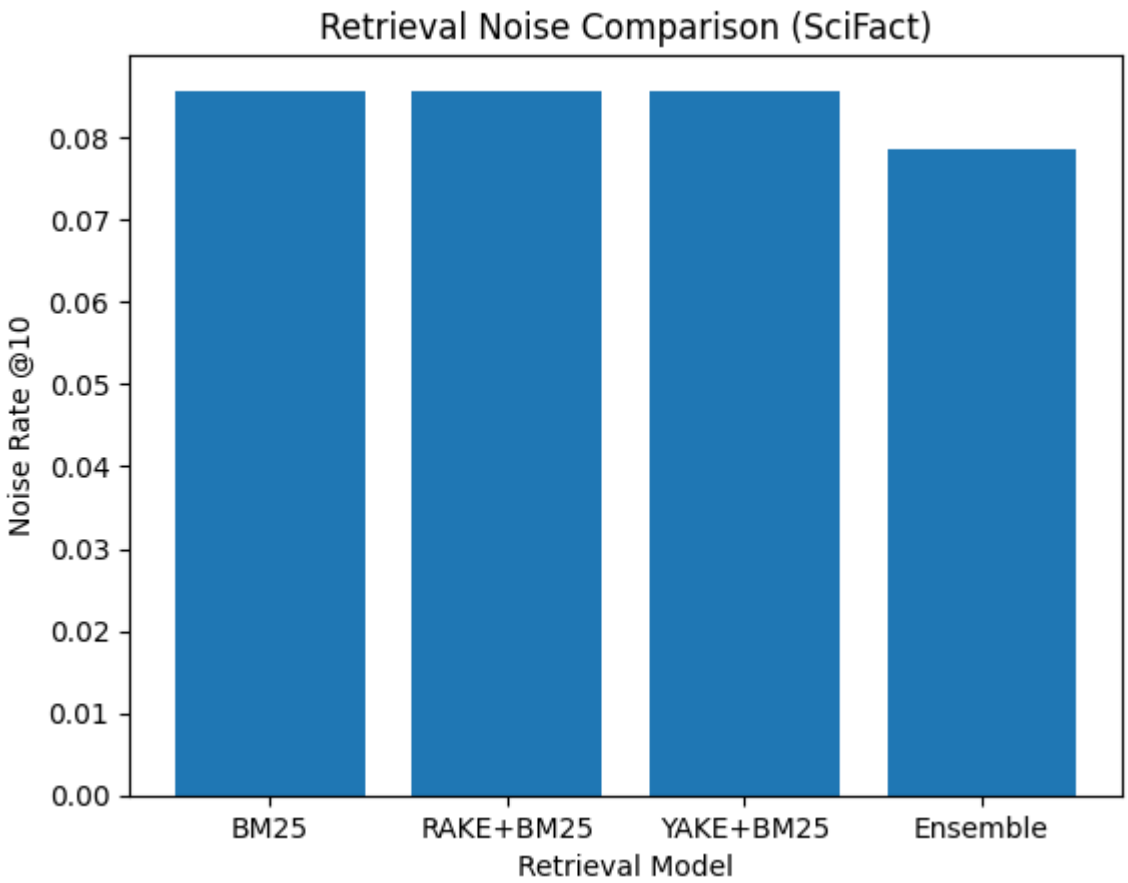

Figure 6: Retrieval noise comparison across retrieval models.

The noise rate represents the proportion of non-relevant documents within the top-k retrieved results.

Lower noise rates indicate that the retrieval system returns more relevant documents and fewer irrelevant ones.

6 Analysis

One of the central motivations of this study is to investigate whether keyword metadata can help

reduce retrieval noise in RAG pipelines.

Retrieval noise occurs when irrelevant documents appear within the top-k retrieved results.

Even when relevant documents are retrieved, the presence of irrelevant documents can reduce the quality of the context

provided to the language model.

To analyze this effect, we measure the noise rate of retrieval results across different retrieval configurations.

Noise rate is defined as the proportion of retrieved documents that are not labeled as relevant within the top-k results.

$$\text{NoiseRate@k} = 1 - (\text{Relevant@k} / k)$$

where Relevant@k denotes the number of relevant documents retrieved within the top-k results.

Figure 6 compares the noise rate across retrieval models on the SciFact dataset.

The results show that keyword-enhanced retrieval models produce slightly lower noise rates compared to the baseline BM25 retriever.

Although the reduction in noise is modest, the results suggest that keyword metadata can provide useful signals

for guiding ranking decisions. By emphasizing documents whose extracted keywords better match the query,

the retrieval system can slightly improve the signal-to-noise ratio within the retrieved candidate set.

These findings suggest that keyword metadata should not replace lexical retrieval signals, but rather function as a lightweight auxiliary feature that complements existing retrieval methods.

7 Conclusion

This paper investigated the role of keyword ensemble metadata as an auxiliary signal in retrieval pipelines

for Retrieval-Augmented Generation systems.

We evaluated keyword metadata generated using RAKE and YAKE on three BEIR benchmark datasets:

SciFact, NFCorpus, and Arguana. Our experiments analyzed keyword representation quality, retrieval performance,

reranking behavior, and parameter sensitivity through alpha-weighted score fusion.

The results show that keyword metadata functions as a weak but useful auxiliary retrieval signal.

Across datasets, small keyword weights ($\alpha \approx 0.05$–0.1) provide the most consistent improvements,

while larger weights degrade retrieval quality.

Additionally, our analysis of retrieval noise indicates that keyword metadata can slightly improve the

signal-to-noise ratio of retrieved documents. Although the improvements are modest, they demonstrate that

keyword-based metadata can provide useful ranking signals when combined with lexical retrieval methods.

Future work will explore integrating keyword metadata with dense retrieval models and adaptive weighting

strategies for metadata signals in hybrid retrieval pipelines.

Keywords: Retrieval-Augmented Generation, Information Retrieval, Keyword Extraction, Metadata Retrieval, Retrieval Noise